# California State University, Long Beach

Mechanical and Aerospace Engineering Department

MAE 300: Engineering Instrumentation and Measurement

**Experiment No. 2**

Calibration of a Load Cell and a Bourdon-Tube-Type Pressure Gage

**Lab Section 19**

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## Abstract:

The objective of this experiment is to introduce students to the calibration procedure of different types of pressure-based measurement tools as well as how to utilize the method of least squares. The Bourdon gauge measured the loads in pounds per square inch (psi), and the load unit cell measured the weight of the loads in pounds. The first instrument that the students used was the load unit cell. Students were given six one kilogram (kg) weights to incrementally load onto the load cell; the students recorded each increment's measurement as the weights were placed onto the load cell as well as when each of the weights were taken off the load cell. The students used a Bourdon pressure gauge to record the measurements of five different weights ranging from 0.5 pounds (lb), 1 lb, and 1.5 lb. The same process for incrementally loading the gauge and recording the data on the load cell was repeated for the Bourdon pressure gauge. Two trials were conducted for each of the measurement apparatuses. The method of least squares was then used to analyze the measurements gathered from the load cell and the Bourdon pressure gauge.

## Introduction:

The Bourdon coil was invented in 1849 by French watchmaker and engineer, Eugene Bourdon. Bourdon's observations of tube coil's tendency to straighten under pressure fueled the creation of the Bourdon pressure gauge. Bourdon's observations became the foundation of the measuring principle of the Bourdon pressure gauge, in which a curved tube with an elliptical cross section would measure pressure based on the deflection of its tip. Bourdon's measuring device is still being utilized in many applications, due to its simplistic operating procedure as well as its accuracy. Although the Bourdon pressure gauge was historically used for steam engines, it is now used in a variety of different industries and engineering fields. Mainly in those involving pressurized systems, such as Heating, Ventilation, and Air Conditioning (HVAC), manufacturing, water treatment, oil drilling, and the automotive industries.

One important thing to keep in mind for the Bourdon pressure gauge, as well as many other measuring apparatuses, is the process of calibration. The Bourdon pressure gauge must be calibrated properly and regularly to maintain accurate pressure readings. It is also important to factor the environment as variable in the calibration process. Over time, thermal stress, wear, improper use, and even environmental exposure can affect the accuracy of the Bourdon pressure gauge as well as many other measurement instruments. Regular calibration is part of the quality control process and is necessary in following regulations and safety standards. Due to the importance of correct calibration and instrument maintenance, statistical analysis such as the method of least squares becomes an invaluable process in analyzing data sets for instrument calibration errors.

## Background & Theory:

Calibration is the process of validating a measuring instrument by comparing its output against a known reference value. Through this comparison, systematic errors and inconsistencies within the instrument can be identified and corrected, ensuring that recorded measurements accurately reflect true physical quantities.

When analyzing the calculated values, the relationship between the input and the output values is expressed using a linear equation in the form of equation (1). Where y is the output, x is the input, $a_0$ is the slope of the line, and $a_1$ is the y-intercept. $a_1$ can also represent the offset error, which is the reading on a measurement apparatus that is shown before adding weight. To find $a_0$, the equation (2) is used and to find $a_1$ equation (3) is used. Equation (1) forms the foundation of linear regression analysis and describes the ideal linear relationship between the two variables, the output, measured values, and the input, or true values. Once the line of best fit is established, the correlation coefficient r can be calculated using equation (6) to assess the strength of the linear relationship between the measured and true values. Therefore, to have a perfect linear relationship between the input and output values, a coefficient of regression (r) should equal $\pm 1$. With the sign of r indicating whether the direction of the variables are the same.

In practice, however, measured data points do not fall perfectly on this line due to factors such as human error, instrument noise, or environmental disturbances. To account for this, the method of least squares is applied to determine the line of best fit which minimizes the total deviation between the predicted and actual data points. To find the best line possible, each individual error is calculated using equation (4). This equation is used to find the distance of each data point from the line of best fit. Where each value is squared to ensure that these distances do not cancel out and punish larger valued errors. Finally, these values are summed up and divided by the number of data points minus 2, this equation (5) gives us the standard error. In addition, the percentage error can be calculated using equation (7) to indicate the percentage error of the output value from the input value.

(1) $y = a_0 + a_i x$

(2) $a_0 = \dfrac{\Sigma x_i \Sigma x_i y_i - \Sigma y_i \Sigma x_i^2}{(\Sigma x_i)^2 - n(\Sigma x_i)^2}$

(3) $a_1 = \dfrac{\Sigma x_i \Sigma y_i - n\Sigma x_i y_i}{(\Sigma x_i)^2 - n\Sigma x_i^2}$

(4) $\varepsilon_i = (y_i - (a_0 + a_1 x_i))^2$

(5) $SE = \sqrt{\dfrac{\Sigma(y_i - (a_0 + a_1 x))^2}{n-2}}$

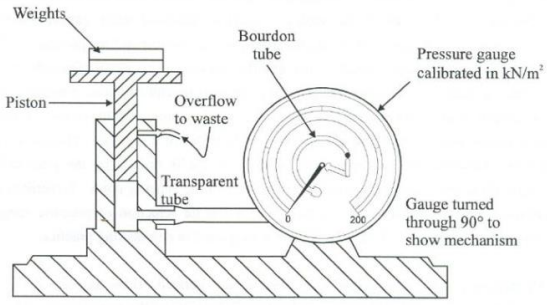
(6) $r = \dfrac{n(\Sigma x_i y_i) - (\Sigma x_i)(\Sigma y_i)}{\sqrt{(n\Sigma x_i{}^2) - (\Sigma x_i)^2)(n\Sigma y_i{}^2) - (\Sigma y_i)^2)}}$

$$(7) \quad \%Error = \left| \frac{input\ mass - |output\ mass|}{input\ mass} \right| \cdot 100$$

## Experimental Procedures:

For this experiment, two apparatuses were used to record pressure. One was an electric compression load cell and the other was a Bourdon pressure gauge. Prior to using and/or not using a measuring device for an extended period, ____it introduces errors in its measurement. To calibrate the Bourdon pressure gauge, its piston is to be removed and put back into the holder that is connected to the transparent tube containing a pressurized fluid. When weight is placed on the piston platform, the pressurized fluid communicates with the pressure gauge. To get a proper mass, the values of the weights and the area of the platform are multiplied.






The electric load cell is calibrated by taring, zeroing, the sensor that is dependent on the applied pressure on the platform, against a known pressure. This device measures pressure with the use of electrical transducers.







For both apparatuses, after calibration, add a known weight and with the addition of each weight, record the pressure—while loading and unloading.

## Raw data:

Figure 1.1: Trial 1 Compression Load Cell Measurements

| Input Data (kg) | Output Data (kg) | | |
|---|---|---|---|
| | Loading | Unloading | Averaged Output Data |
| 0 | 0 | 0.002 | 0.000 |
| 2.2 | 2.140 | 2.139 | 2.289 |
| 4.4 | 4.267 | 4.268 | 9.106 |
| 6.6 | 6.382 | 6.383 | 20.368 |
| 8.8 | 8.496 | 8.498 | 36.100 |
| 11 | 10.602 | 10.6 | 56.191 |
| 13.2 | 12.703 | 12.703 | 80.683 |

Figure 1.2: Trial 2 Compression Load Cell Measurements

| Input Data (kg) | Output Data (kg) | | |
|---|---|---|---|
| | Loading | Unloading | Averaged Output Data |
| 0 | 0.000 | 0.000 | 0.000 |
| 2.2 | 2.142 | 2.139 | 2.291 |
| 4.4 | 4.236 | 4.235 | 8.970 |
| 6.6 | 6.307 | 6.306 | 19.886 |
| 8.8 | 8.360 | 8.358 | 34.936 |
| 11 | 10.410 | 10.41 | 54.184 |
| 13.2 | 12.476 | 12.476 | 77.825 |

Figure 1.3: Trial 1 Bourdon Pressure Gauge Measurements

| Input Data (lb) | Output Data (lb) | | |
|---|---|---|---|
| | Loading | Unloading | Averaged Output Data |
| 0 | 3.1232 | 3.3184 | 5.1820 |
| 0.5 | 3.5624 | 3.8552 | 6.8669 |
| 1.5 | 4.5872 | 4.8312 | 11.0808 |
| 2.5 | 5.5144 | 5.856 | 16.1462 |
| 3.5 | 6.4416 | 6.9296 | 22.3189 |
| 5 | 7.9544 | 7.9544 | 31.6362 |

Figure 1.4: Trial 2 Bourdon Pressure Gauge Measurements

| Input Data (lb) | Output Data (lb) | | |
|---|---|---|---|
| | Loading | Unloading | Averaged Output Data |
| 0 | 3.2696 | 3.3184 | 5.4249 |
| 0.5 | 3.66 | 3.7088 | 6.7871 |
| 1.5 | 4.7824 | 4.7824 | 11.4357 |
| 2.5 | 5.6608 | 5.7096 | 16.1605 |

| | | | |
|---|---|---|---|
| 3.5 | 6.6856 | 6.7344 | 22.5118 |
| 5 | 8.052 | 8.052 | 32.4174 |

## Sample Calculations and Results:

Table 1.5: Loading Compression Cell Calibration Equation Values

| n | Input, x (lb) | Output, y (lb) | $xy$ | $x^2$ | $y^2$ | $(y-(a_0+a_1x))^2$ |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0.000 | 0.000 | 0.000 | 0.00020 |
| 2 | 2.2 | 2.140 | 4.708 | 4.840 | 4.580 | 0.00006 |
| 3 | 4.4 | 4.267 | 18.775 | 19.360 | 18.207 | 0.00029 |
| 4 | 6.6 | 6.382 | 42.121 | 43.560 | 40.730 | 0.00020 |
| 5 | 8.8 | 8.496 | 74.765 | 77.440 | 72.182 | 0.00010 |
| 6 | 11 | 10.602 | 116.622 | 121.000 | 112.402 | 0.00000 |
| 7 | 13.2 | 12.703 | 167.680 | 174.240 | 161.366 | 0.00036 |

Figure 1.1: Compression Cell Loading Graph

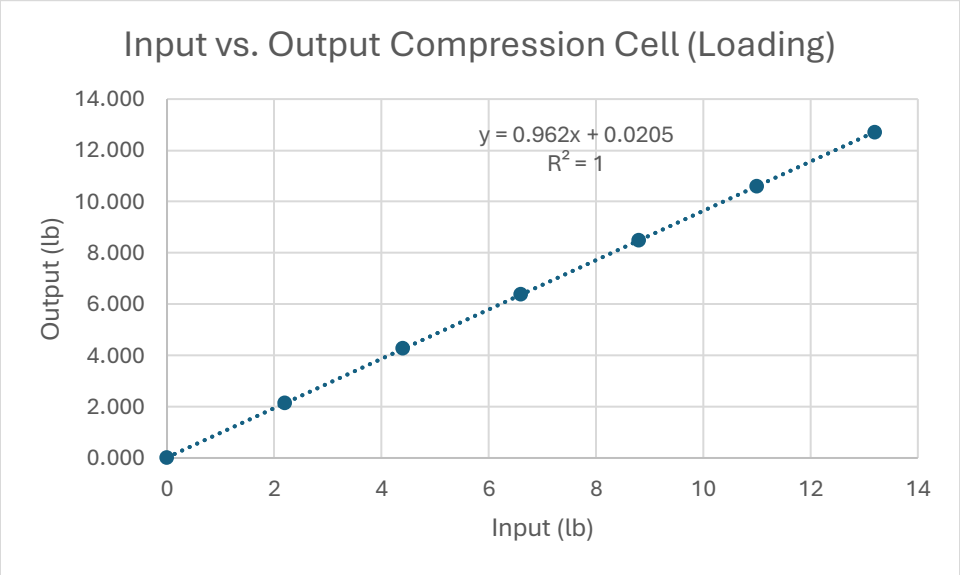



Table 1.6: Unloading Compression Cell Calibration Equation Values

| n | Input, x (lb) | Output, y (lb) | $xy$ | $x^2$ | $y^2$ | $(y-(a_0+a_1x))^2$ |
|---|---|---|---|---|---|---|
| 1 | 0 | 0.002 | 0.000 | 0.000 | 0.000 | 0.00016 |
| 2 | 2.2 | 2.139 | 4.706 | 4.840 | 4.575 | 0.00004 |
| 3 | 4.4 | 4.268 | 18.779 | 19.360 | 18.216 | 0.00031 |
| 4 | 6.6 | 6.383 | 42.128 | 43.560 | 40.743 | 0.00022 |
| 5 | 8.8 | 8.498 | 74.782 | 77.440 | 72.216 | 0.00014 |
| 6 | 11 | 10.600 | 116.600 | 121.000 | 112.360 | 0.00002 |
| 7 | 13.2 | 12.703 | 167.680 | 174.240 | 161.366 | 0.00035 |

Figure 1.2: Compression Cell Unloading Graph

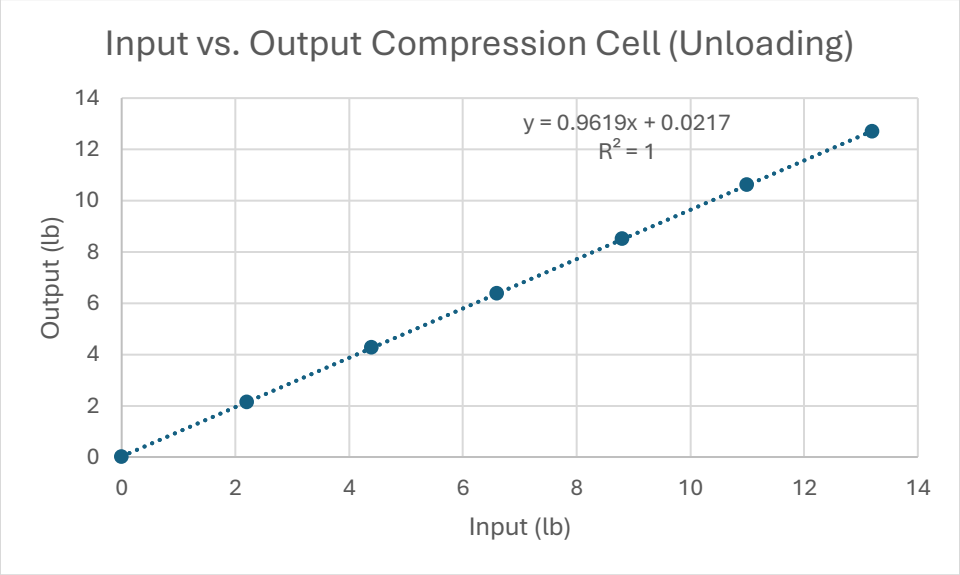



Figure 1.3: Compression Cell Average Graph

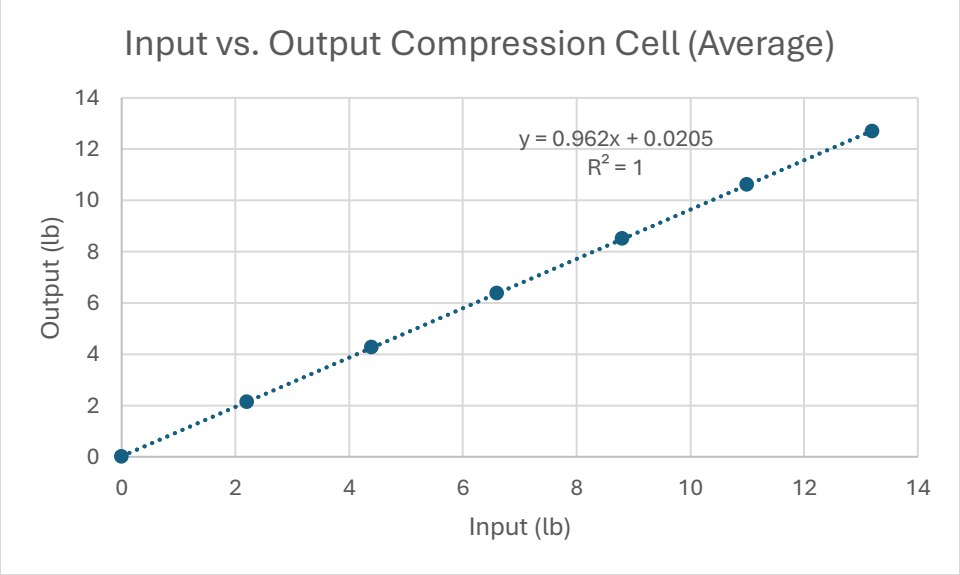



Table 1.7: Loading/Unloading Compression Cell Value Comparison

| | $r$ | $a_0$ | $a_1$ | $SE$ | $y = a_0 + a_i x$ |
|---|---|---|---|---|---|
| **Loading** | 1.000 | 0.0205 | 0.962 | 0.0150 | $y = 0.0205 + 0.962x \pm 0.005$ |
| **Unloading** | 1.000 | 0.0217 | 0.9619 | 0.0151 | $y = 0.0217 + 0.9619x \pm 0.005$ |
| **Average** | 1.000 | 0.0205 | 0.962 | 0.0150 | $y = 0.0205 + 0.962x \pm 0.005$ |

Table 1.8: 1.11: Percentage Error for Loading Compression Cell

| **Input (lb)** | **Output (lb)** | **%Error (%)** |
|---|---|---|
| 0 | 0.000 | 0 |

| | | |
|---|---|---|
| 2.2 | 2.140 | 2.727 |
| 4.4 | 4.267 | 3.023 |
| 6.6 | 6.382 | 3.303 |
| 8.8 | 8.496 | 3.455 |
| 11 | 10.602 | 3.618 |
| 13.2 | 12.703 | 3.765 |

Table: 1.9: Percentage Error for Unloading Compression Cell

| **Input (lb)** | **Output (lb)** | **%Error (%)** |
|---|---|---|
| 0 | 0.002 | 0 |
| 2.2 | 2.139 | 2.773 |
| 4.4 | 4.268 | 3.000 |
| 6.6 | 6.383 | 3.288 |
| 8.8 | 8.498 | 3.432 |
| 11 | 10.600 | 3.636 |
| 13.2 | 12.703 | 3.765 |

Figure 1.4: Percentage Error Compression Cell Graph

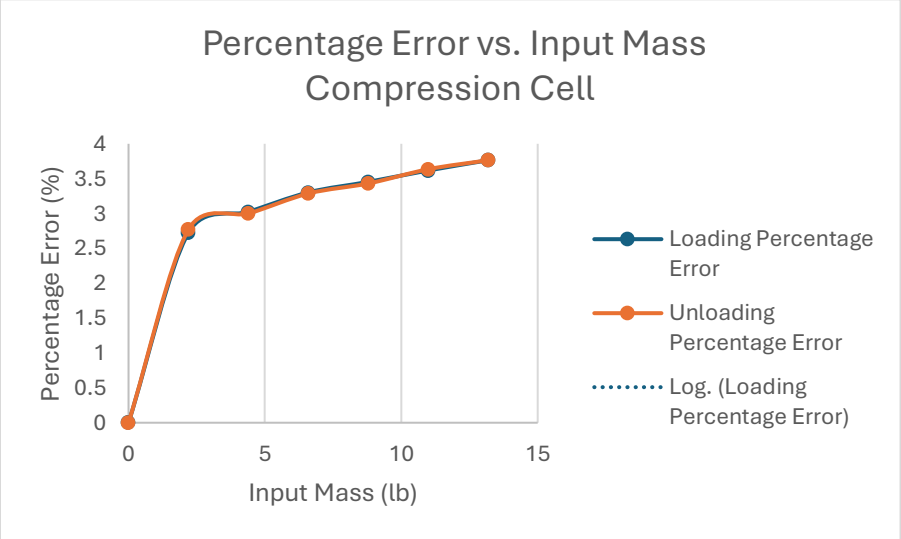



Table 1.10: Loading Bourdon Pressure Gauge Calibration Equation Values

| **n** | **Input, x (kg)** | **Output, y (kg)** | $xy$ | $x^2$ | $y^2$ | $(y-(a_0+a_1x))^2$ |
|---|---|---|---|---|---|---|
| 1 | 2.2 | 3.1232 | 6.871 | 4.840 | 9.754 | 0.00025 |
| 2 | 2.7 | 3.5624 | 9.618 | 7.290 | 12.691 | 0.00074 |
| 3 | 3.7 | 4.5872 | 16.973 | 13.690 | 21.042 | 0.00109 |
| 4 | 4.7 | 5.5144 | 25.918 | 22.090 | 30.409 | 0.00002 |
| 5 | 5.7 | 6.4416 | 36.717 | 32.490 | 41.494 | 0.00174 |
| 6 | 7.2 | 7.9544 | 57.272 | 51.840 | 63.272 | 0.00058 |

Figure 1.5: Bourdon Pressure Gauge Loading Graph

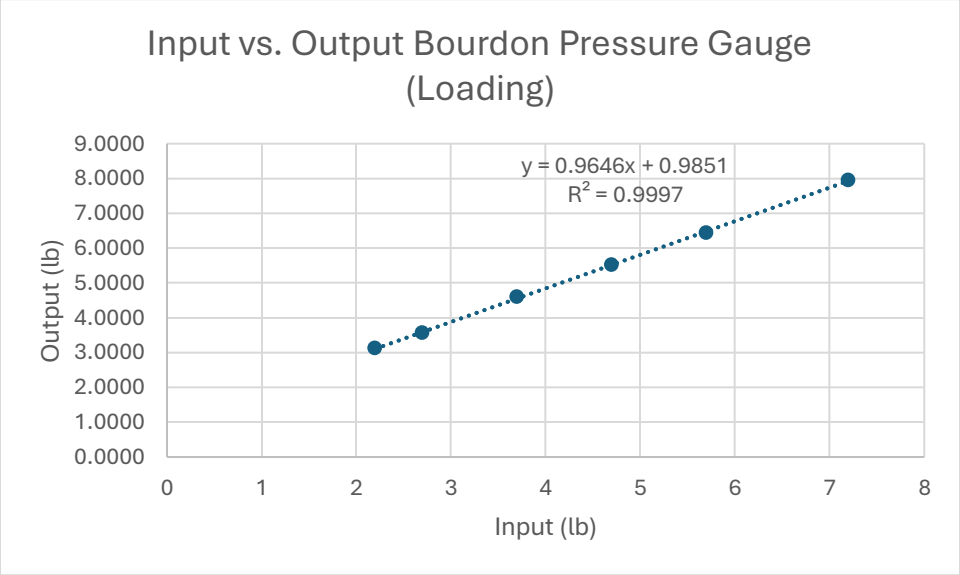



Table 1.11: Unloading Bourdon Pressure Gauge Calibration Equation Values

| **n** | **Input, x (lb)** | **Output, y (lb)** | $xy$ | $x^2$ | $y^2$ | $(y-(a_0+a_1x))^2$ |
|---|---|---|---|---|---|---|
| 1 | 2.2 | 3.3184 | 7.300 | 4.840 | 11.012 | 0.00751 |
| 2 | 2.7 | 3.8552 | 10.409 | 7.290 | 14.863 | 0.00055 |
| 3 | 3.7 | 4.8312 | 17.875 | 13.690 | 23.340 | 0.00003 |
| 4 | 4.7 | 5.856 | 27.523 | 22.090 | 34.293 | 0.00685 |
| 5 | 5.7 | 6.9296 | 39.499 | 32.490 | 48.019 | 0.04373 |
| 6 | 7.2 | 7.9544 | 57.272 | 51.840 | 63.272 | 0.03497 |

Figure 1.6: Bourdon Pressure Gauge Unloading Graph

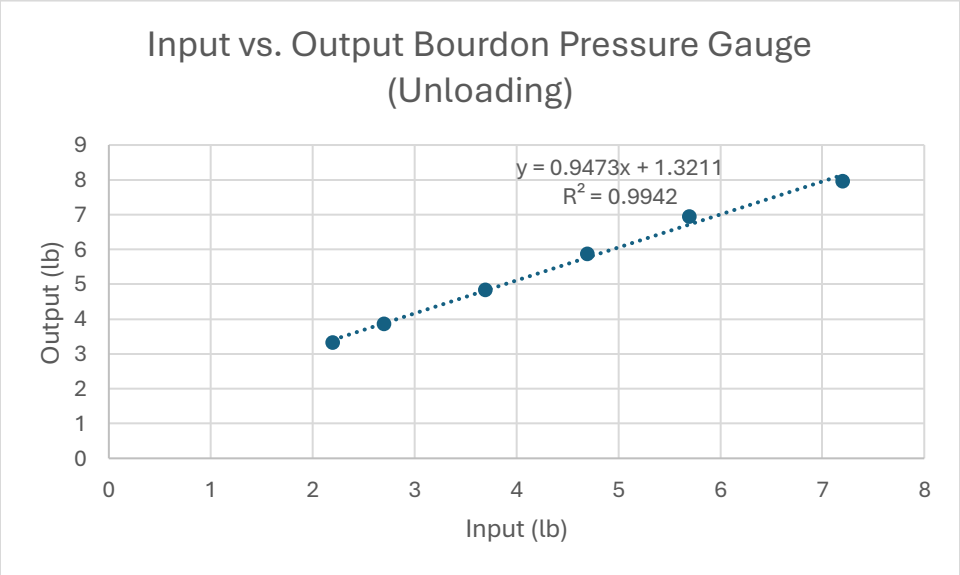


Figure 1.7: Bourdon Pressure Gauge Average Graph

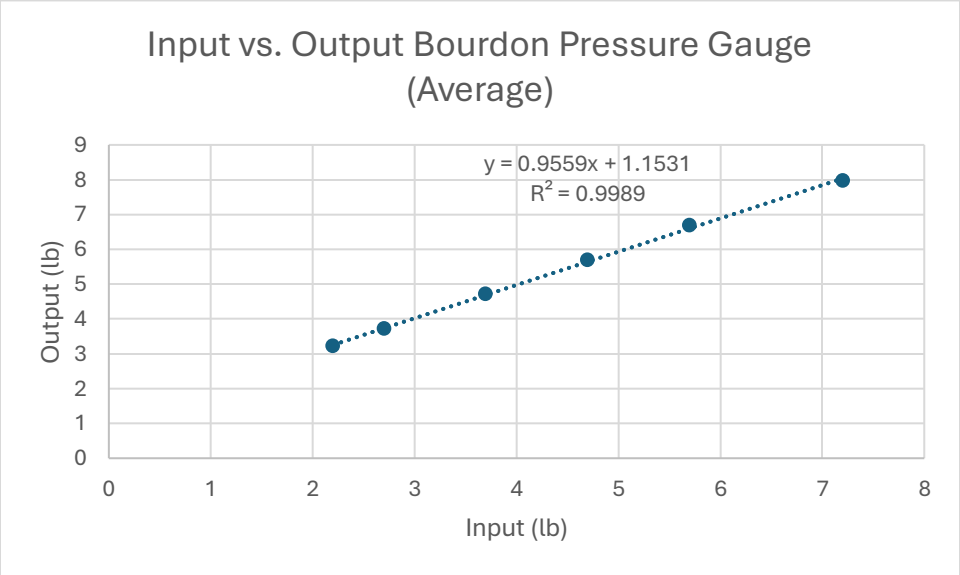



Table 1.12: Loading/Unloading Bourdon Pressure Gauge Value Comparison

| | $r$ | $a_0$ | $a_1$ | $SE$ | $y = a_0 + a_i x$ |
|---|---|---|---|---|---|
| **Loading** | 0.9997 | 0.9851 | 0.9646 | 0.0333 | $y = 0.9851 + 0.9646x \pm 0.50$ |
| **Unloading** | 0.9942 | 1.3211 | 0.9473 | 0.1530 | $y = 0.9473 + 1.3211x \pm 0.50$ |
| **Average** | 0.9989 | 1.1531 | 0.9559 | 0.0660 | $y = 0.9559 + 1.1531x \pm 0.50$ |

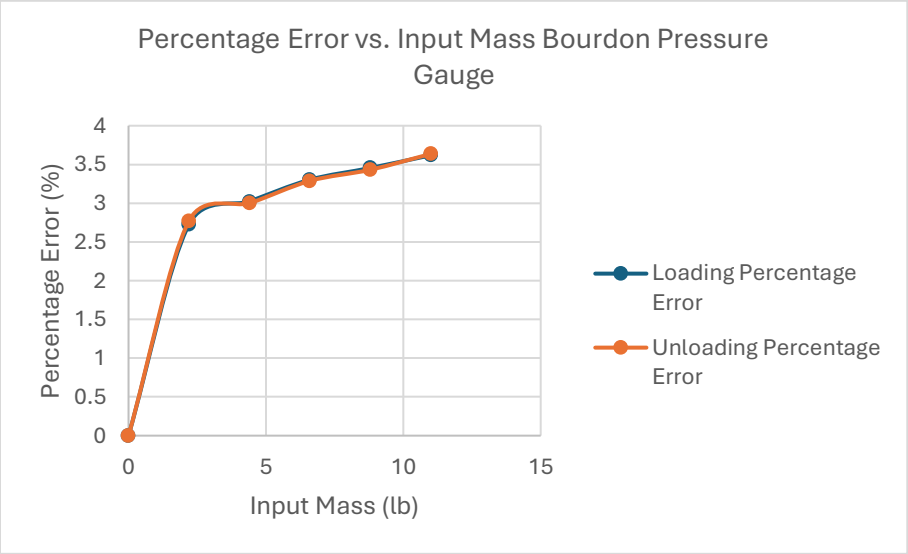
Table: 1.13: Percentage Error for Loading Bourdon Pressure Gauge

| **Input** | **Output** | **%Error** |
|---|---|---|
| 0 | 0.000 | 0 |
| 2.2 | 2.140 | 2.727 |
| 4.4 | 4.267 | 3.023 |
| 6.6 | 6.382 | 3.303 |
| 8.8 | 8.496 | 3.455 |
| 11 | 10.602 | 3.618 |

Table: 1.14: Percentage Error for Unloading Bourdon Pressure Gauge

| **Input** | **Output** | **%Error** |
|---|---|---|
| 0 | 0.002 | 0 |
| 2.2 | 2.139 | 2.773 |
| 4.4 | 4.268 | 3.000 |
| 6.6 | 6.383 | 3.288 |
| 8.8 | 8.498 | 3.432 |
| 11 | 10.600 | 3.636 |

Figure 1.8: Percentage Error Bourdon Pressure Gauge Graph




## Discussion of results:

Part I: Compression Load Cell:

In the first part of the experimental calculations, the loading measurements from the compression load cell were used to develop the calibration equation (5) for the loading, unloading and average output measurements. The loading calibration equation is $y = 0.0205 + 0.962x \pm 0.005$. The unloading calibration equation is $y = 0.0217 + 0.9619x \pm 0.005$. And the average output calibration equation is $y = 0.0205 + 0.962x \pm 0.005$. The correlation coefficient was r = +1.000 for the loading, unloading, and average output measurement data sets. Having a correlation coefficient of +1.000 indicates that each of the data sets have a strong linear relationship and because $r > 0$, the variables move in the same direction. This correlation coefficient indicates a minimal non-linearity error. When comparing the values of $a_0$ and $a_1$ for the loading, unloading and average data sets, they are close in value. With the values of $a_0$ and $a_1$ being the same for the loading and average data sets, indicating a minor hysteresis error between the loading and unloading of the compression load cell. In addition, because the values of $a_0$ are relatively the same, this means that there is also minimal zero-shift error. And comparing the values of $a_1$ show that there is minimal sensitivity error.

The standard error, which indicates how much the output measurements vary from the input load are close in value, ranging from 0.0150 to 0.0151. The smaller value for the standard error means that the data are relatively accurate to the input load values and that each of the data sets fall within a narrow range of variation from the input load. Analyzing the calculated percentage error values for the loading and unloading show that the measurements were relatively accurate to the actual input value of the weights. The value of the percentage error ranges from 0% to 3.765% for the loading and unloading data set. The values of the percentage errors for both the loading and unloading data sets are relatively close in value, which means that

the values for the input measurements were close to the unloading measurements. This can also be observed from the minimal hysteresis error.

Part II: Bourdon Pressure Gauge:

In the second part of the experiment, the students used a Bourdon pressure gauge to gather loading and unloading measurements for five weights ranging from 0.5 lb to 1.5 lb. The same method of least squares and calculations were conducted for these measurements. The resulting calibration equation for the loading data set is $y = 0.9851 + 0.9646x \pm 0.50$ and r = +0.997. The calibration equation for the unloading data set is $y = 0.9473 + 1.3211x \pm 0.50$ and r = +0.9942. The calibration equation for the average data set is $y = 0.9559 + 1.1531x \pm 0.50$ and r = +0.9989. The $a_1$ values for each of the calibration equations are significant in that the $a_1$ values represent the initial reading of the Bourdon pressure gauge before adding the plunger, platform, and weights. The difference between the value of $a_1$ for the loading versus unloading data sets indicate that as the weights were incrementally added onto the platform of the Bourdon gauge, the initial pressure readings shifted. The value of $a_1$ for the loading data set was 0.9851 while the unloading value of $a_1$ was 0.9473, so there was a difference of 0.0378 between these values. Although there is a slight variation in the values of the correlation coefficients from each of the data sets, they fall within a narrow range. In addition, they are all positive values and are close to 1, which means that there is a strong linear relationship between each of the output data sets and the input load. The positive value of r indicates that variables move in the same direction. Having a correlation coefficient so close to +1 means that there is minimal non-linearity errors within the data sets. The unloading data set has a slightly larger non-linearity error due to it having the smallest r value. Comparing the values of $a_0$, there is a zero shift error that occurs between the graph of the loading and unloading graphs. The largest difference between the values of $a_0$ is between the loading and unloading graphs with a difference of 0.336. However, comparing the values of $a_1$, there is only a variation of 0.0173 so there is minimal sensitivity error.

The standard error, which indicates how much the output measurements vary from the input load are close in value, range from 0.0333 to 0.1530. The smaller values for the standard error for the loading and average data set mean that the values are relatively accurate to the input load values. However, the higher standard error for the unloading direction means that this data set had less accuracy relative to the loading direction. Analyzing the calculated percentage error values for the loading and unloading show that the measurements were relatively accurate to the actual input value of the weights. The value of the percentage error ranges from 0% to 3.681% for the loading data set and 0% to 3.636% for the unloading data set. The values of the percentage errors for both the loading and unloading data sets are relatively close in value, which means that the values for the input measurements were close to the output measurements. There is a slight difference between the values of the percentage error for the unloading and loading. This indicates that percent error favors the unloading input direction because of the smaller maximum percentage error value.

## Conclusions and Recommendations:

The process of calibrating and utilizing the Bourdon pressure gauge and the compression load cell was insightful. Analyzing the differences between the loading and unloading output values for the different measurements provided students with greater knowledge of the importance of instrument calibration and the analysis of error propagation. In addition, the use of the method of least squares to calculate the calibration equations, correlation coefficients, and standard errors, provided students with valuable experience. From this experiment, students were able to identify different types of errors that propagated within their data, such as hysteresis errors, non-linearity errors, zero shift errors, and calibration errors. Identifying which calculations and values the students analyzed to indicate the different types of errors further emphasized the importance of statistical analysis in measurement collection and experimentation.

All in all, the values that were collected and calculated showcased minimal errors between the loading and unloading graphs for the Bourdon pressure gauge and the compression load cell. However, in comparing the standard error values for the Bourdon pressure gauge, the students were able to identify the unloading data set as having a greater variation from the input data.

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